

Exercise 1: Implementing the Singleton Pattern

```
1 class Logger {  
2  
3     // Private static instance  
4     private static Logger instance;  
5  
6     // Private constructor  
7     private Logger() {  
8         System.out.println("Logger Instance Created");  
9     }  
10  
11  
12     public static Logger getInstance() {  
13         if (instance == null) {  
14             instance = new Logger();  
15         }  
16         return instance;  
17     }  
18  
19     public void log(String message) {  
20         System.out.println("LOG: " + message);  
21     }  
22 }  
23  
24
```

```
25 public class SingletonTest {  
26  
27     public static void main(String[] args) {  
28  
29         Logger logger1 = Logger.getInstance();  
30         Logger logger2 = Logger.getInstance();  
31  
32         logger1.log("First Message");  
33         logger2.log("Second Message");  
34  
35         if (logger1 == logger2) {  
36             System.out.println("Only one instance exists.");  
37         } else {  
38             System.out.println("Multiple instances exist.");  
39         }  
40     }  
41 }
```

Output:

```
Logger Instance Created  
LOG: First Message  
LOG: Second Message  
Only one instance exists.
```

Compiled and executed in 1.818 sec(s)